

DILI — Overview: Direct (Intrinsic) vs Idiosyncratic



1. DILI banner

WHAT YOU SEE

Drug-Induced Liver Injury sign over the goat farm

WHAT IT ENCODES

DILI = liver injury caused by drugs, herbals/supplements, or toxins. It is a leading cause of acute liver failure in the US and the #1 reason drugs are withdrawn from the market. The core dichotomy to master is intrinsic/direct (predictable, dose-dependent) vs idiosyncratic (unpredictable, host-dependent).

BOARD TRAP

BOARD TRAP: don't forget herbal/dietary supplements (e.g., green tea extract, anabolic 'bodybuilding' supplements, kava) as DILI causes — a patient on no prescription drugs can still have DILI from OTC supplements.

2. Red dragon (Direct/Intrinsic)

WHAT YOU SEE

Fire-breathing red dragon labeled DIRECT TOXICITY (INTRINSIC)

WHAT IT ENCODES

Intrinsic/direct DILI is dose-dependent, predictable, reproducible, and short-latency (hours to days); the toxin or its reactive metabolite directly injures hepatocytes, so it is reproducible in animal models. Acetaminophen is the prototype.

BOARD TRAP

WRONG to call intrinsic DILI 'unpredictable' or 'host-dependent' — predictability and dose dependence are its defining features and the key discriminator from idiosyncratic injury.

3. APAP bullseye target

WHAT YOU SEE

Bullseye target labeled PREDICTABLE next to APAP (acetaminophen)

WHAT IT ENCODES

Acetaminophen is the classic intrinsic hepatotoxin and the #1 cause of acute liver failure in the US/West. Toxicity occurs above ~150 mg/kg or ~10 g acutely (lower with chronic alcohol/fasting). Management is the Rumack-Matthew nomogram (4-hour level) and N-acetylcysteine.

BOARD TRAP

WRONG to wait for the 4-hour acetaminophen level before giving NAC when timing is uncertain or ingestion is staggered — start NAC empirically; delay is the fatal error, and NAC is most effective within 8 hours.

4. Dose-dependent gauge + stopwatch

WHAT YOU SEE

Gauge labeled DOSE DEPENDENT and a stopwatch for SHORT LATENCY (hours)

WHAT IT ENCODES

Intrinsic injury scales with dose and appears within hours, contrasting sharply with idiosyncratic injury's dose-independence and variable 5–90 day latency. The short, dose-related course is what makes intrinsic toxins predictable and reproducible.

BOARD TRAP

WRONG to use a quick onset to argue AGAINST drug causality in general — only intrinsic toxins act within hours; idiosyncratic DILI's weeks-to-months latency means a 'normal LFT last month' does not exclude the drug.

5. NAPQI dragon fire

WHAT YOU SEE

Flame labeled NAPQI shooting from dragon toward the goat

WHAT IT ENCODES

Acetaminophen toxicity is mediated by the reactive metabolite NAPQI (formed via CYP2E1). NAPQI depletes hepatic glutathione and binds proteins, causing centrilobular (zone 3) necrosis. NAC works by replenishing glutathione so NAPQI is detoxified.

BOARD TRAP

WRONG to think chronic alcohol use is protective — it induces CYP2E1, generating MORE NAPQI, so alcoholics develop acetaminophen toxicity at LOWER doses (and glutathione stores are depleted by fasting/malnutrition).

6. Direct examples placard

WHAT YOU SEE

List: acetaminophen, Amanita mushroom, carbon tetrachloride

WHAT IT ENCODES

Classic intrinsic/direct hepatotoxins: acetaminophen, Amanita phalloides (foraged 'death cap' mushroom; amatoxin causes biphasic GI then hepatic failure), carbon tetrachloride, and yellow phosphorus. All are dose-dependent and predictable.

BOARD TRAP

WRONG to manage Amanita poisoning as routine gastroenteritis — after a deceptive symptom-free interval, fulminant hepatic failure ensues; recognize the foraged-mushroom history and consider early transfer for possible transplant.

7. Crowned goat (Diagnosis of Exclusion)

WHAT YOU SEE

Crowned goat holding scales with PRESUMPTIVE DIAGNOSIS OF EXCLUSION

WHAT IT ENCODES

DILI is a clinical diagnosis of EXCLUSION — there is no confirmatory test. Establish a temporal relationship to drug exposure, document improvement on dechallenge, and systematically rule out other causes. RUCAM/Roussel-Uclaf scoring formalizes causality.

BOARD TRAP

WRONG to rely on liver biopsy 'to confirm DILI' — histology is supportive but nonspecific; the diagnosis is clinical and the priority is excluding viral, autoimmune, and obstructive disease.

8. Exclude first list

WHAT YOU SEE

Red X list: viral hepatitis, autoimmune hepatitis, biliary disease, other liver diseases

WHAT IT ENCODES

Before diagnosing DILI you MUST exclude viral hepatitis (HAV IgM, HBsAg/anti-HBc IgM, HCV RNA, HEV in the right setting), autoimmune hepatitis (ANA/ASMA, IgG), biliary obstruction (RUQ ultrasound/MRCP), and other liver disease (Wilson, hemochromatosis, ischemic hepatitis).

BOARD TRAP

WRONG to anchor on the drug and skip imaging/serologies — missing acute viral hepatitis or a stone obstructing the common bile duct (treatable, potentially fatal) is the classic exclusion trap.

9. Masked bandit (Idiosyncratic)

WHAT YOU SEE

Masked thief labeled IDIOSYNCRATIC (EXTRINSIC) rolling dice

WHAT IT ENCODES

Idiosyncratic DILI is unpredictable, NOT clearly dose-dependent, and host-dependent (immune or metabolic susceptibility, often HLA-linked), with variable latency. It accounts for most prescription-drug DILI and the majority of drug-related acute liver failure outside acetaminophen.

BOARD TRAP

WRONG to assume a small or 'safe' dose excludes idiosyncratic DILI — it is dose-independent, so it can occur at standard or low doses, unlike the strict dose threshold of intrinsic toxins.

10. Only 1/1000–10000 dice

WHAT YOU SEE

Dice/sign reading ONLY 1/10³–10⁵ PATIENTS

WHAT IT ENCODES

Idiosyncratic reactions are rare — roughly 1 in 1,000 to 1 in 100,000 exposed patients. This rarity is precisely why they are not detected in pre-marketing clinical trials and only surface in post-marketing surveillance, and why they trigger drug withdrawals.

BOARD TRAP

WRONG to dismiss a drug as the cause because the reaction is 'too rare' — population rarity does not lower causality in the individual patient once a temporal relationship and dechallenge response are established.

11. Latency 5-90 days calendar

WHAT YOU SEE

Calendar marked LATENCY 5–90 DAYS

WHAT IT ENCODES

Idiosyncratic DILI typically appears 5–90 days (weeks to a few months) after starting the drug. Immune/hypersensitivity types tend to be earlier (1–6 weeks) and recur rapidly on rechallenge; metabolic types can be later and more insidious.

BOARD TRAP

WRONG to exclude a drug because LFTs were normal a few weeks earlier — the long, variable latency means injury can emerge well after the drug is started, or even (e.g., amox-clav) after it is stopped.

12. Fever/rash/eosinophilia thermometer

WHAT YOU SEE

Thermometer + spotted petri dish: SOMETIMES fever, rash, eosinophilia

WHAT IT ENCODES

A subset of idiosyncratic DILI is hypersensitivity (allergic/immune) type with fever, rash, and eosinophilia — the spectrum of DRESS (Drug Reaction with Eosinophilia and Systemic Symptoms). Classic with anticonvulsants, allopurinol, and sulfonamides. These features are NOT always present.

BOARD TRAP

WRONG to require fever/rash/eosinophilia before diagnosing idiosyncratic DILI — metabolic-type injury (e.g., isoniazid) has none of these, so their absence does not exclude DILI.

13. Idiosyncratic examples placard

WHAT YOU SEE

List: INH, nitrofurantoin, amoxicillin-clavulanate, minocycline

WHAT IT ENCODES

Classic idiosyncratic offenders: amoxicillin-clavulanate (the #1 cause of idiosyncratic DILI, cholestatic/mixed, delayed onset), isoniazid (hepatocellular), nitrofurantoin (chronic AIH-like + pulmonary fibrosis), and minocycline (drug-induced AIH/lupus-like).

BOARD TRAP

WRONG to pick acetaminophen as the answer to 'most common cause of idiosyncratic DILI' — APAP is the most common INTRINSIC hepatotoxin; amoxicillin-clavulanate is the most common IDIOSYNCRATIC cause.

14. Board trap: keep giving drug

WHAT YOU SEE

Angry doctor still handing OFFENDING DRUG to a sick yellow goat

WHAT IT ENCODES

The single most important step in DILI management is to STOP the offending drug. Continuing it despite clinical illness or jaundice drives progression to acute liver failure; most cases improve after prompt dechallenge.

BOARD TRAP

WRONG answer: continuing the drug 'to finish the course' in a symptomatic or jaundiced patient — this is the fatal trap; jaundice in hepatocellular injury signals Hy's Law (>10% mortality) and mandates stopping.